SYNAPTIX Research Division

Project Overview: Edge ML for Predictive Maintenance

Project ID: SYNAPTIX-PM-2023-2025 | Period: March 2023 – June 2025

1. Objective

Develop and validate multi-modal machine learning models that run at the edge to predict equipment failures in industrial sensor networks. Success criteria: (a) F1 ≥ 0.85, (b) failure lead time ≥ 10 days, (c) estimated energy savings ≥ 15% versus reactive maintenance baselines.

2. Hypothesis

Combining vibration, temperature, and acoustic features in a quantized neural network deployed on edge devices will detect impending failures earlier and more reliably than single-modality approaches, while reducing unnecessary continuous high-power sensing and thereby lowering energy consumption.

3. Key Results (Summary)

Metric	Target	Achieved
Best F1 Score	≥ 0.85	0.92
Precision / Recall	—	0.94 / 0.91
Lead Time (days)	≥ 10	9.8
Energy Saving %	≥ 15%	17.5%
Total Samples	—	15,450

4. Methodology Snapshot

Ten controlled experiments (EXP-PM-001 through EXP-PM-010) progressively added sensor modalities and model improvements. Full methodology is documented in Methodology.md. Experimental results are recorded in Experimental_Results.xlsx and the detailed feature dataset in Research_Dataset.csv.

5. Notable Anomaly

EXP-PM-006 (acoustic-only) underperformed (F1 0.71, FPR 0.14). Root cause: insufficient labeled acoustic failure events. This serves as a planted limitation for analysis testing. See Limitations_and_Future_Work.md.

6. Document Map

- Research_Project_Overview.pdf (this document)
- Literature_Review.docx — fictional internal prior work synthesis
- Experimental_Results.xlsx — full experiment table
- Research_Dataset.csv — 500-sample feature extract from EXP-PM-009
- Methodology.md, Research_Findings.json, Research_Chart.png

Synthetic research project for SYNAPTIX COMMAND testing. All data fictional.